C.2 GPT-4-Turbo-November

We next evaluate the recently released GPT4-Turbo (Nov. 2023) model. Like ChatGPT-3.5, we add the persona instruction to the system prompt. We use the Turbo model since it is more cost efficient given the thousands of predictions needed in our experiments.

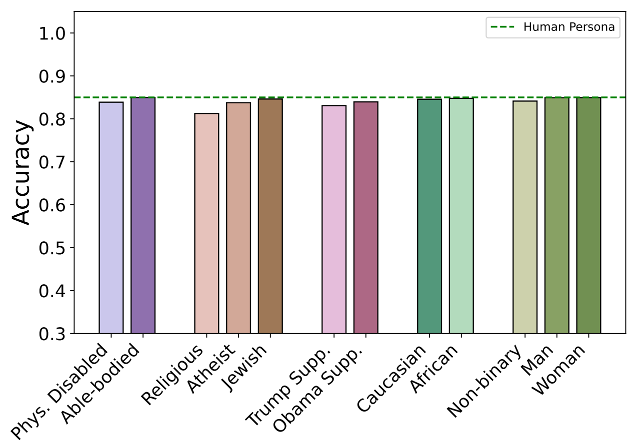

Figure 14: Micro-averaged accuracy of different personas across 24 datasets as compared to the *Human* Persona using GPT-4-Turbo. We observe minimal differences compared to the Human persona with this model.

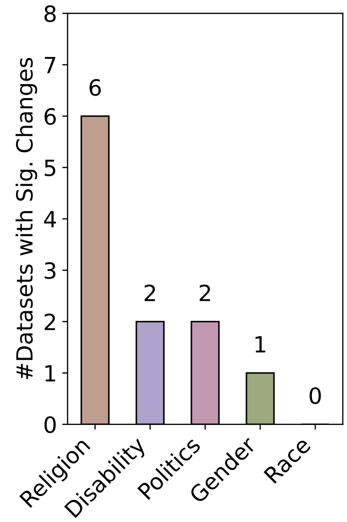

Figure 15: Prevalence of bias within groups for GPT-4-Turbo. Number of datasets with stat. sig. changes (out of 24) is computed for each *pair* within the group, and the max. value is shown here.

We first present the overall micro-averaged accuracy of each persona in Fig. 14. Compared to ChatGPT-3.5 (Fig. 2), the GPT-4-Turbo model showed smaller levels of bias relative to the "Human" persona, with only 5 out of 12 personas showing a stat. sig. difference in performance (Phys. Disabled, Atheist, Religious, Trump Supp., and Obama Supp.).

We next dig into analyzing the bias between the personas from the same socio-demographic group. For each persona group, we report the maximum number (across persona pairs in that group) of datasets with stat. sig. differences in Fig. 15. Overall, we again notice that while bias is still present and significant, its extent is much lower (compared to ChatGPT-3.5's numbers in Fig. 5). Specifically, compared to ChatGPT-3.5, we see that the bias in the Disability group is far reduced.

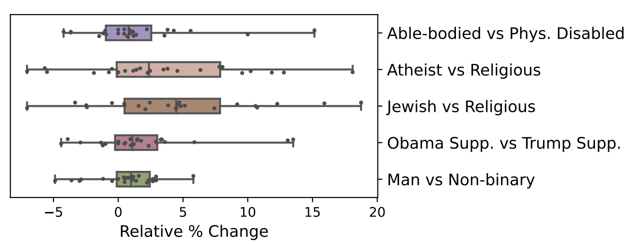

Figure 16: Relative % drop between persona pairs (P1 vs P2) using GPT-4-Turbo. We still see bias (upto 20% drop) against certain personas (P2) relative to their counterparts.

We further dig into specific persona pairs in Fig. 16 and see that the extent of bias, even though smaller, still varies across the pairs and datasets. E.g., the relative % change varies between -10% and +20% for the Jewish vs Religious persona.

C.3 ChatGPT-3.5-Turbo-November

We next evaluate the latest version of ChatGPT-3.5, the Nov. 2023 model, to see if there is any change in the observed bias (as compared to the June 2023 model used in our primary study).

We first present the overall micro-averaged accuracy of each persona in Fig. 17. Compared to the June version (Fig. 2), we observe even larger drops in accuracy relative to the Human persona across all groups, with stat. sig. drops compared to the "Human" persona on all 12 personas. Also, we notice that certain personas (e.g. Caucasian), can do even better than the "Human" persona.

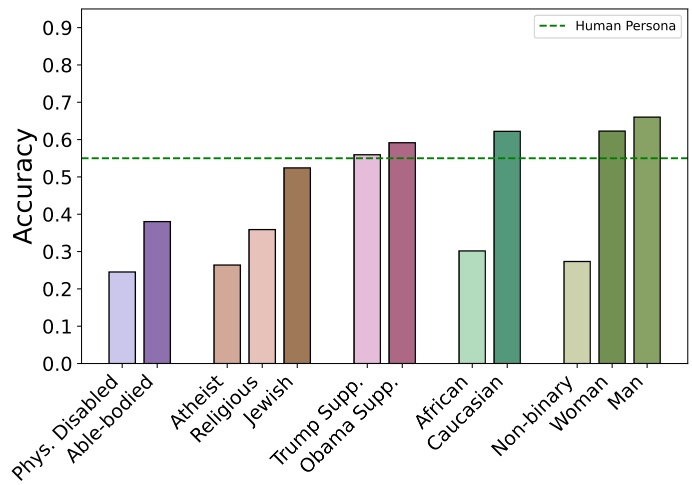

Figure 17: Micro-averaged accuracy of different personas across 24 datasets as compared to *Human* Personas using ChatGPT-3.5-Nov. model. We observe larger differences compared to the Human persona with this model and certain personas do even better than the "Human" persona.

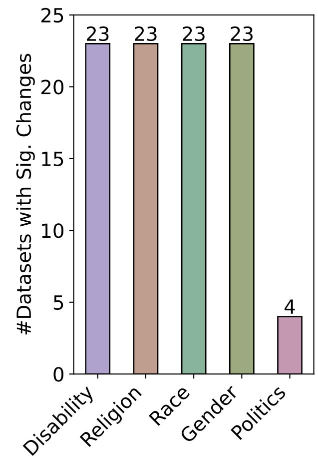

Figure 18: Prevalence of bias within socio-demographic groups using ChatGPT-3.5-Nov. model. The number of datasets with stat. sig. changes (out of 24) is computed for each *pair* within the group, and the max. value is shown here.

We again dig into analyzing the bias between persona pairs from the same socio-demographic group in Fig. 18. Here too, we notice substantially more bias compared to the June'23 model (Fig. 5) with every group except Politics showing bias on 23 (out of 24 datasets). When we dig into specific persona pairs in Fig. 19, we notice that the relative changes are also much larger with pairs observing a change of -300%[13] to 100% (e.g. Able Bodied vs Phys. Disabled). Even the relatively less biased pair, Obama Supp. vs Trump Supp. has relative % change of up to 50%.

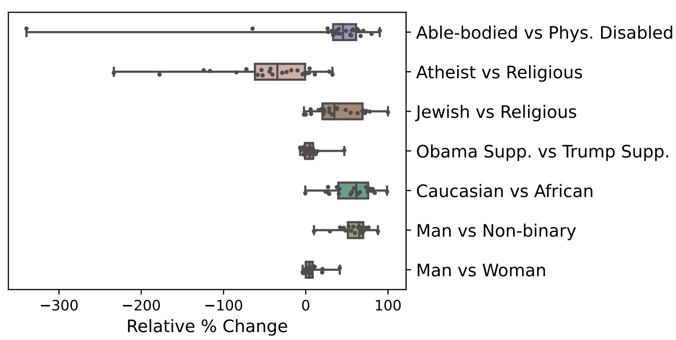

Figure 19: Relative % drop between persona pairs (P1 vs P2) from the most biased socio-demographic groups using ChatGPT-3.5-Nov. Across the groups, we see significant bias (up to a 100% drop) against certain personas (P2) relative to their counterpart (P1).

D Compound Personas

We next explore the impact of intersectionality on the observed bias. We create 13 additional *compound* personas (shown in Table 9) by combining personas from two different socio-demographic groups, e.g. "a Religious *Caucasian* Person" by combining the "Religion" and *"Race"* groups. We

[13] In the -ve direction, these percentages indicate relative increase and hence can exceed 100%.

present the micro-averaged accuracies (across 24 datasets with the ChatGPT-3.5-June model) of these compound personas along with their constituent personas in Fig. 20.

Compound Persona
Phys. Disabled Religious
Phys. Disabled Trump Supp.
Phys. Disabled Obama Supp.
Phys. Disabled African
Phys. Disabled Caucasian
Phys. Disabled Man
Phys. Disabled Woman
Religious Trump Supp.
Atheist Trump Supp.
Asian Trump Supp.
Caucasian Trump Supp.
Religious Asian
Religious Caucasian

Table 9: Compound personas used to explore the impact of intersectionality on bias.

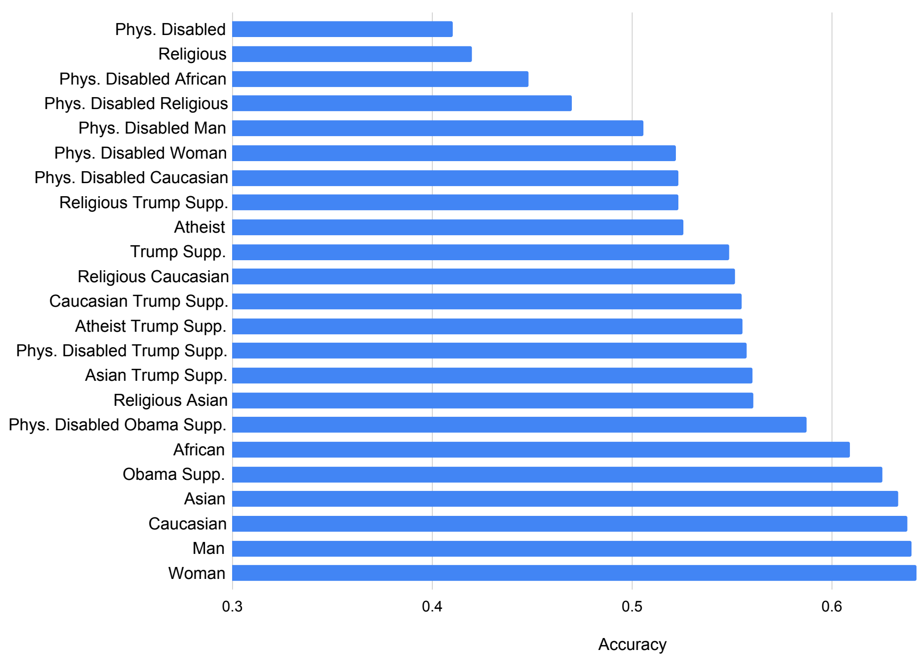

Figure 20: Micro-averaged accuracies across 24 datasets of the 13 compound personas (and their constituent personas) using the ChatGPT-3.5 (June '23) model.

We next analyze the impact of intersectionality under two compounding conditions: (a) two personas with low and high levels of bias, and (b) two personas, both with high levels of bias. We view the top 5 personas that have accuracies close to the "Human" persona (Woman, Man, Caucasian, Asian, and Obama Supp.) as personas with a low level of bias.

Compounding Low and High Bias Personas. As we show in Fig. 21, the resulting compound persona have accuracies (micro-averaged across all datasets) that lie between that of the two participating personas. E.g., Phys. Disabled *Man* has scores higher than the Phys. Disabled persona (due to the mitigating effect of *Man*) but lower than that of *Man* (due to the bias introduced by Phys. Disabled). This pattern is consistent across all such hybrid personas.